Speech Synthesis

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Outline

- Basic introduction to Speech Synthesis
- Components of TTS
- Traditional Methods
- Advanced TTS Systems
- Developing TTS Systems
- Future Directions

Methods

- Formant Speech Synthesis (1960s-1980s): Klatt synthesizer
- Concatenative Speech Synthesis (1990s - Early 2000s): Festival
- Statistical Parametric Speech Synthesis
  - HMM based Speech Synthesis (Early 2000s - Mid 2010s): HTS
  - DNN Based Speech Synthesis (2013s -2017s): Merlin
- Advanced Speech synthesis
  - Deep neural networks (Transformer, RNN, CNNs) (2017 - present): Tacotron 2, WaveNet
  - Architectures like flow/Diffusion based models: (2020-present) Grad-TTS (2021), DiffSinger (2022), StyleTTS2 (2023)
  - LLM Based models (2022 - present): VALL-E (2023), Bark (2023)

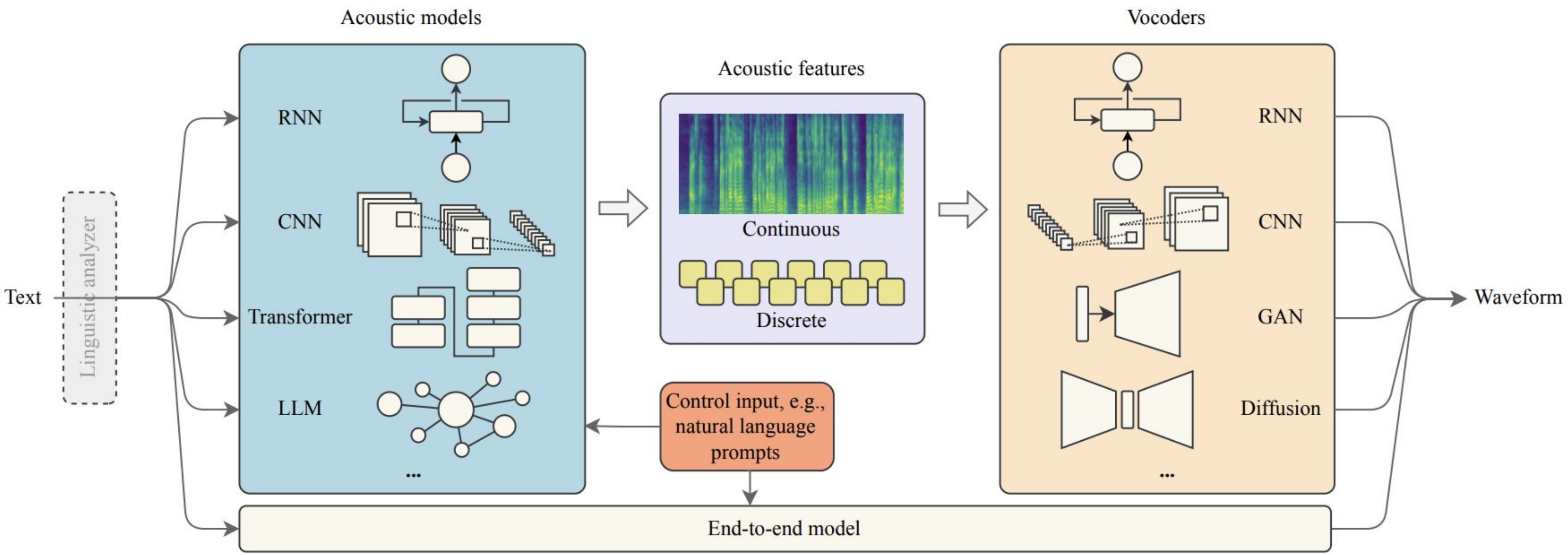
Advanced Speech Synthesis Systems

Advance Synthesis System

- Given text input, synthesize speech corresponding to text
- Two steps:
  - Phoneme-sequence to acoustic-feature representation (Acoustic Model)
    - **Tacotron2**, DeepVoice3, Transformer, FastSpeech, Diffusion based
  - Acoustic-feature to waveform (vocoder)
    - **WaveNet**, WaveRNN, WaveGlow, LPCNet, Parallel WaveNet, Hifi-GAN
- E2E models
  - Combining the acoustic model + vocoder
    - F5-TTS, StyleTTS2
- Tools: ESPNet, Coqui TTS, FastSpeech2, VITS, Speech Brain
- Advantage: Good quality and more control in-terms of style and prosody

(eg: https://google.github.io/tacotron/publications/global_style_tokens/index.html)

Cascaded Models



Xie, Tianxin, et al. "Towards Controllable Speech Synthesis in the Era of Large Language Models: A Survey." *arXiv preprint arXiv:2412.06602* (2024).

Acoustic Model

- Parametric Models
- RNN-based models
- CNN-based models
- Transformer-based models
- LLM-based models: VALL-E, Seed TTS, Base TTS

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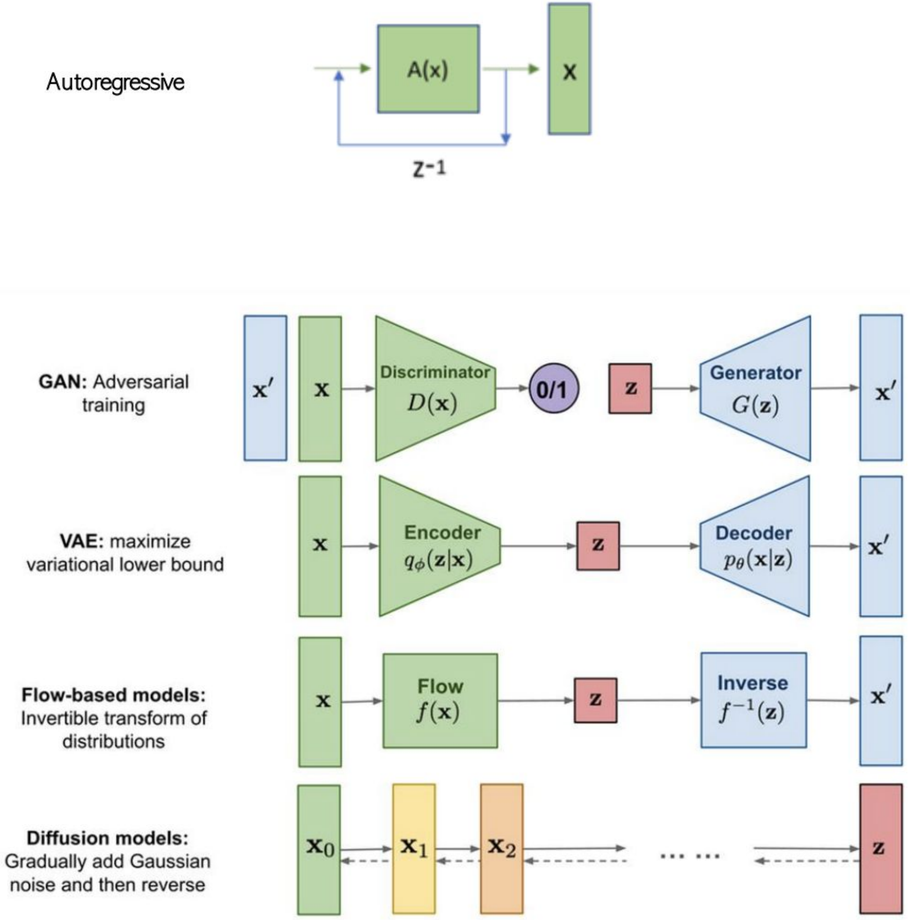
Vocoder



- RNN-based: WaveRNN
- CNN-based: WaveNet
- GAN-based: Parallel WaveGAN, HiFi-GAN
- Diffusion-based: WaveGrad, DiffWave
- Other vocoders: Flow based, VAE based

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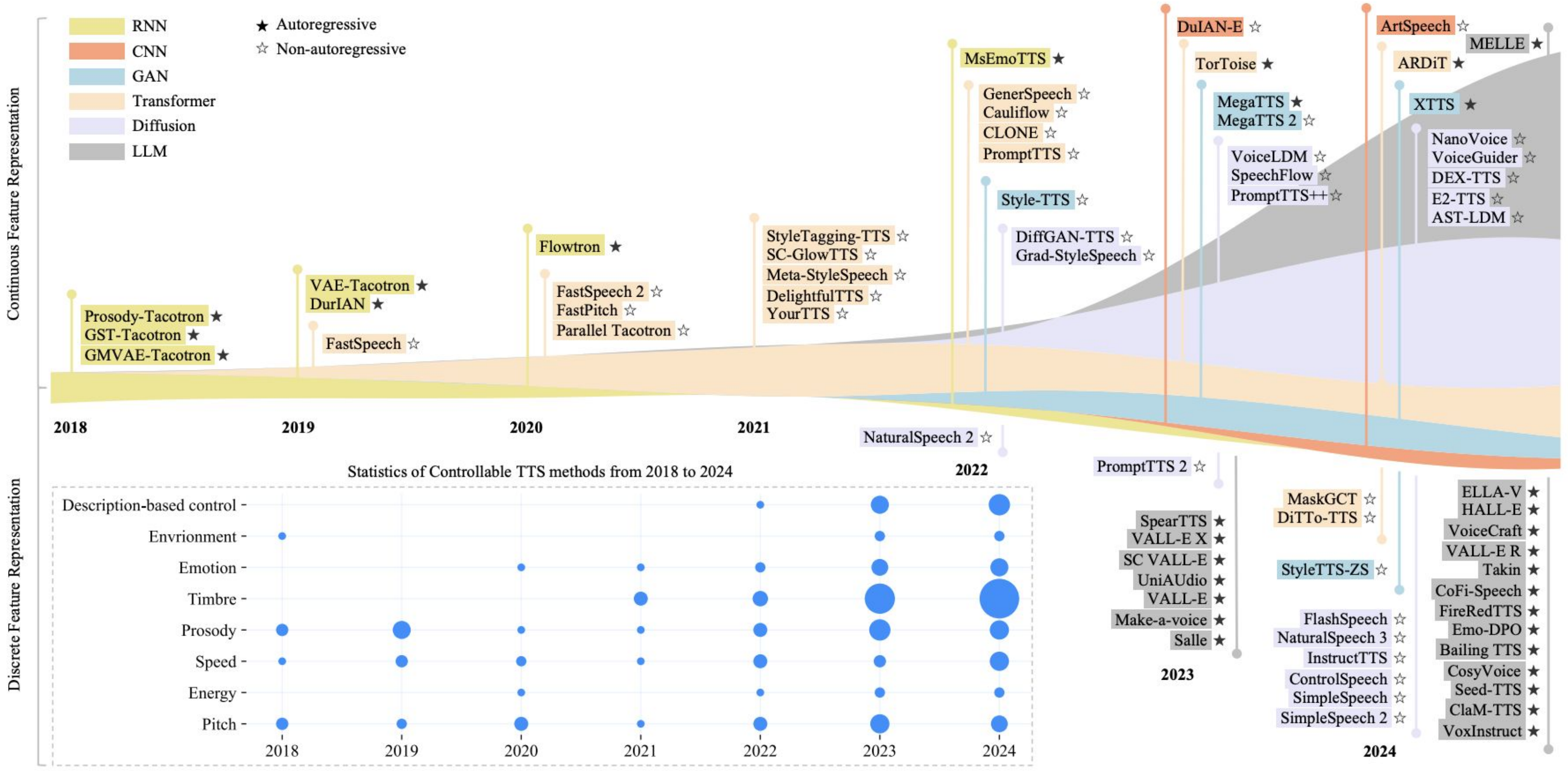
Recent methods



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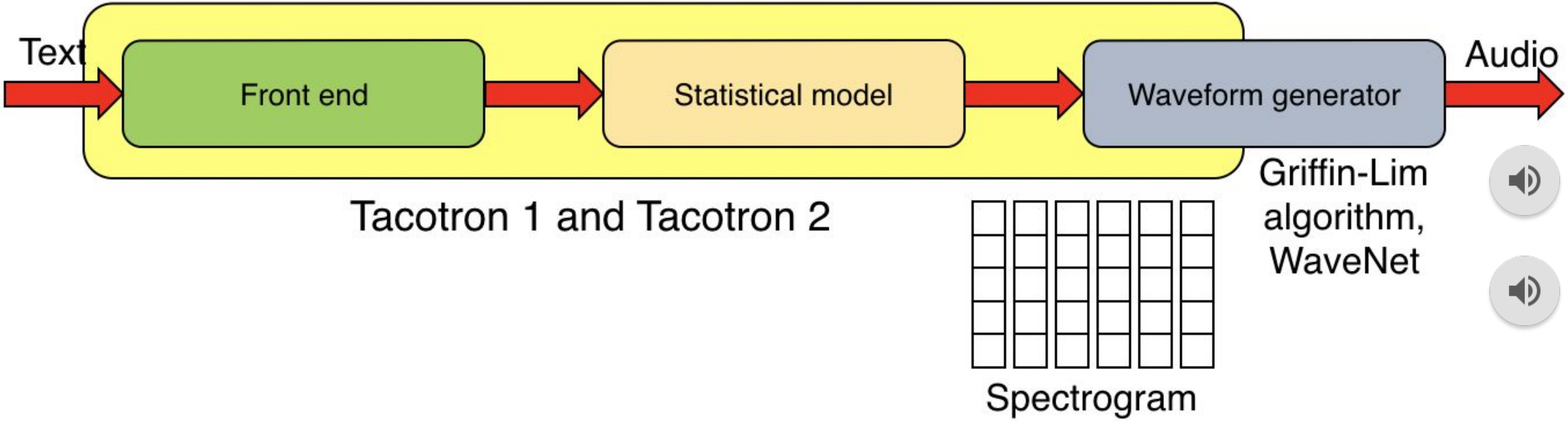
Overview of Tacotron:



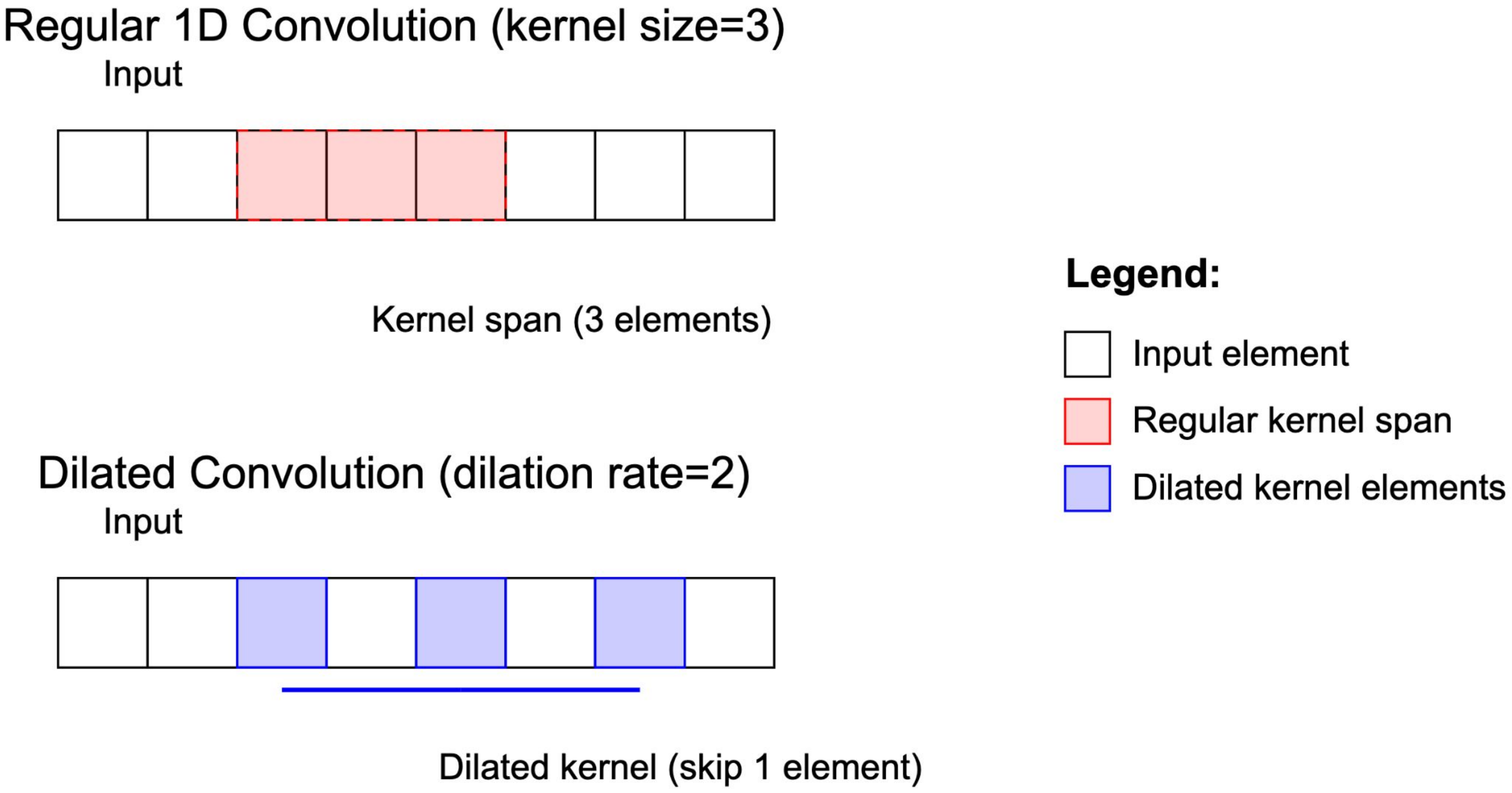
- Tacotron (1, 2) take characters as input and produces spectrogram frames, which are converted to waveforms.
- Tacotron 1 produces linear-scale spectrograms and uses the Griffin-Lim algorithm to synthesize waveform from the predicted spectrogram.
- Tacotron 2 produces mel-scaled spectrograms, which are used as local conditioning to a WaveNet vocoder.
- In contrast to Tacotron 1, Tacotron 2 uses simpler building blocks, using vanilla LSTM and convolutional layers in the encoder and decoder instead of “CBHG” stacks and GRU recurrent layers.

WavNet

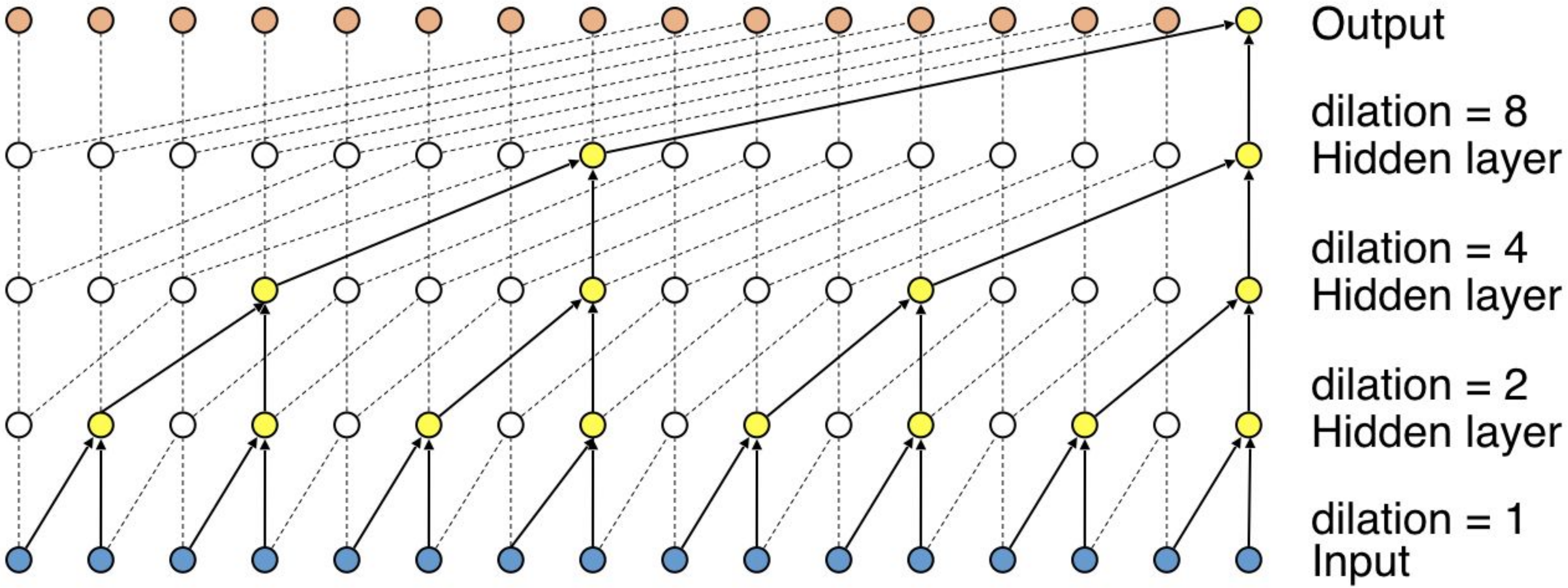
- In September 2016, DeepMind presented WaveNet.
- Wavenet out-performed the best TTS systems (parametric and concatenative) in Mean Opinion Scores (MOS).
- Wavenet Architecture consists of
  - Dilated Causal Convolutions: Kernel elements are spaced apart by a dilation rate
  - Stacked Architecture
  - Residual and Skip Connections
  - Gated Activation Units

van den Oord, Aaron; Dieleman, Sander; Zen, Heiga; Simonyan, Karen; Vinyals, Oriol; Graves, Alex; Kalchbrenner, Nal; Senior, Andrew; Kavukcuoglu, Koray, WaveNet: A Generative Model for Raw Audio, arXiv:1609.03499, 2016

1D convolution vs Dilated convolution



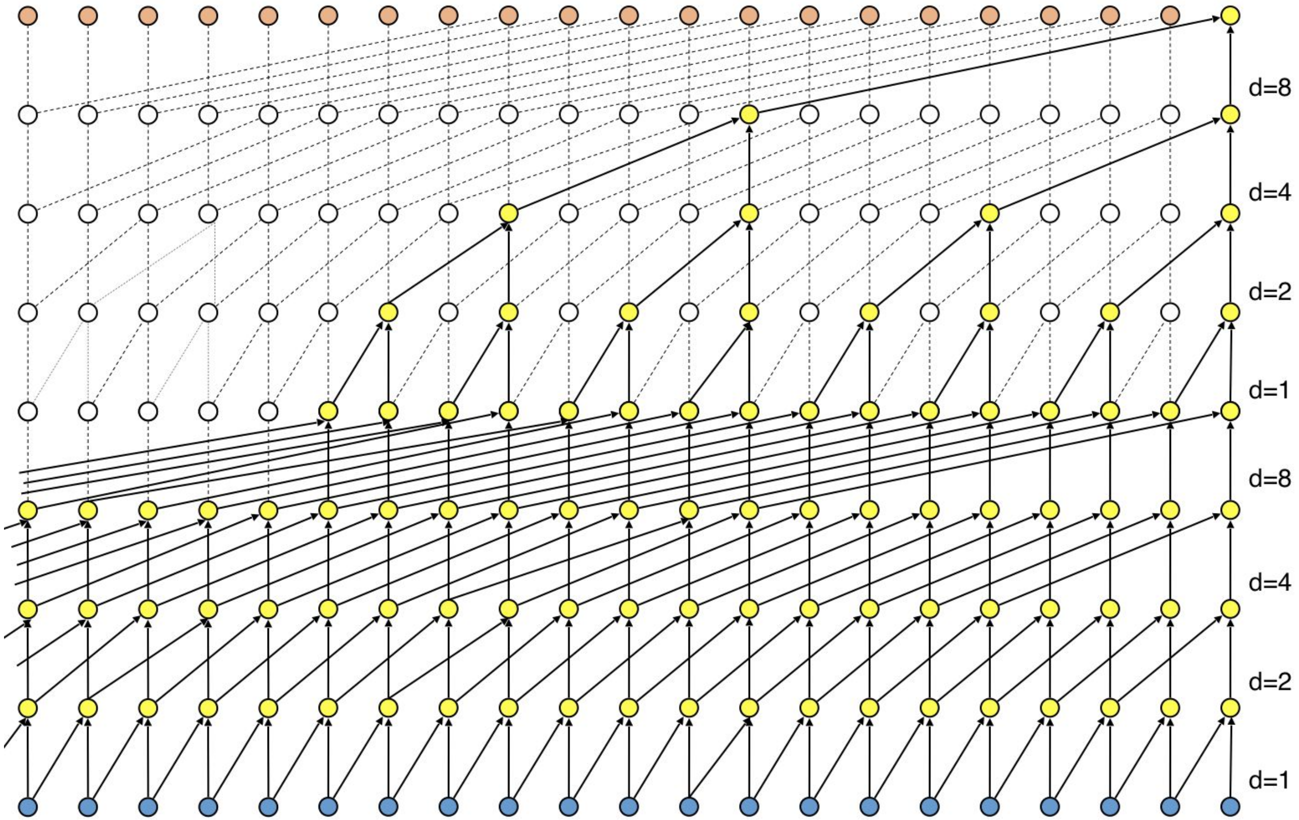
WaveNet: Architecture



- Stacked dilated convolutions enable very large receptive fields with just a few layers.
- The receptive field of the above example is (8+4+2+1) + 1 = 16
- In WaveNet, the dilation is doubled for every layer up to a certain point and then repeated: 1, 2, 4, ..., 512, 1, 2, 4, ..., 512, 1, 2, 4, ..., 512, 1, 2, 4, …, 512, 1, 2, 4, …, 512

WaveNet: Stacking multiple blocks

- Example with dilations 1,2,4,8,1,2,4,8



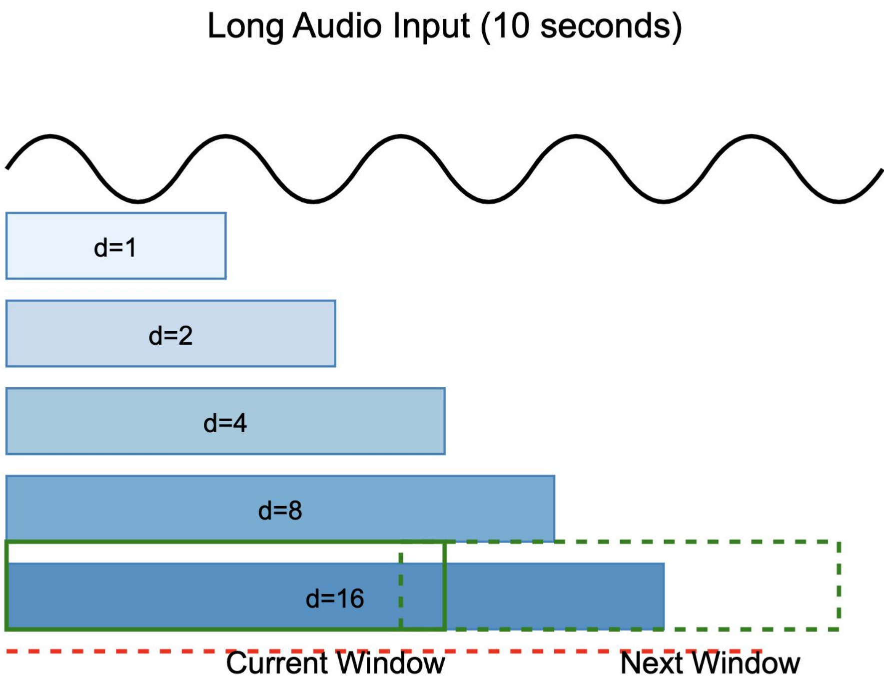
Audio processing

Layer 1: dilation = 1 -> receptive field = 2

Layer 2: dilation = 2 -> receptive field = 4

Layer 3: dilation = 4 -> receptive field = 8

Layer 4: dilation = 8 -> receptive field = 16



For 10 seconds of audio at 16 kHz sampling rate:

Input is divided into chunks with appropriate overlap

Total samples = 10 seconds × 16000 Hz = 160,000 samples

Receptive field = 512 samples ≈ 32ms of audio

Receptive field = 2048 samples ≈ 128ms of audio

All the above chunks can be processed parallelly

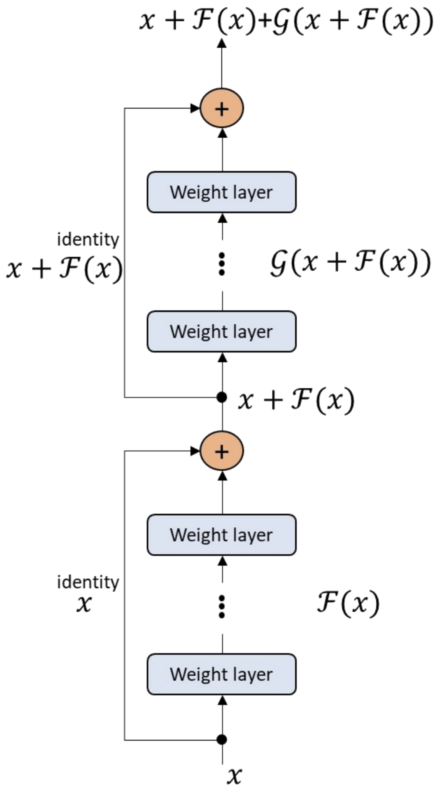
Wavenet – Residual and skip connections

Residual Connection:

- Adds input directly to the processed output
- Helps with gradient flow
- Enables training of very deep networks
- Formula: output = input + processed_features

Skip Connection:

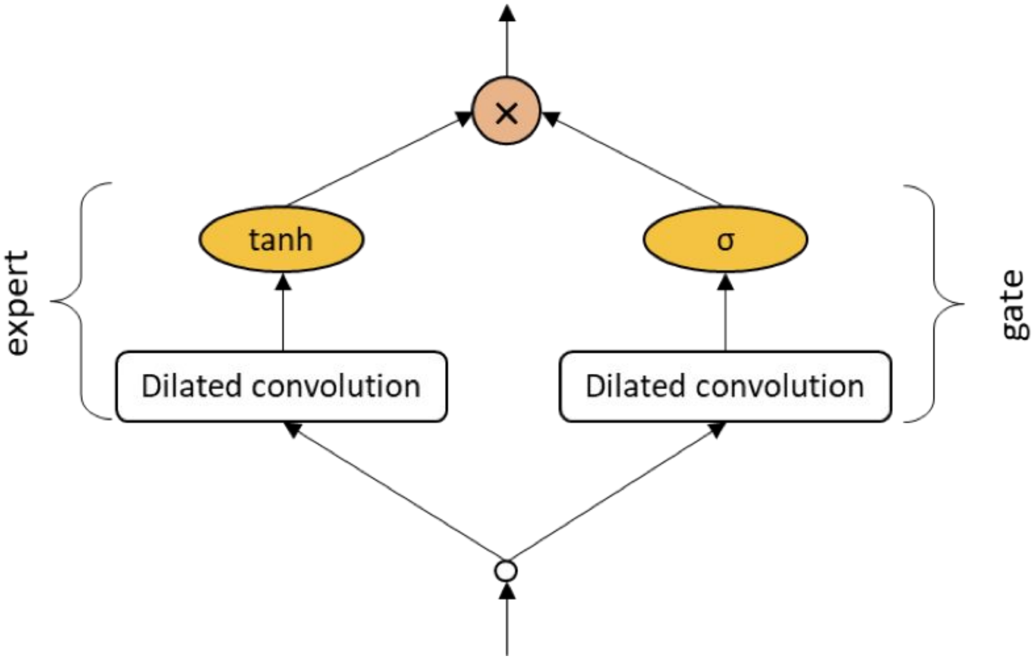
- Taken after the 1x1 convolution
- Bypasses multiple layers
- All skip connections are summed at the end
- Helps capture features at different time scales



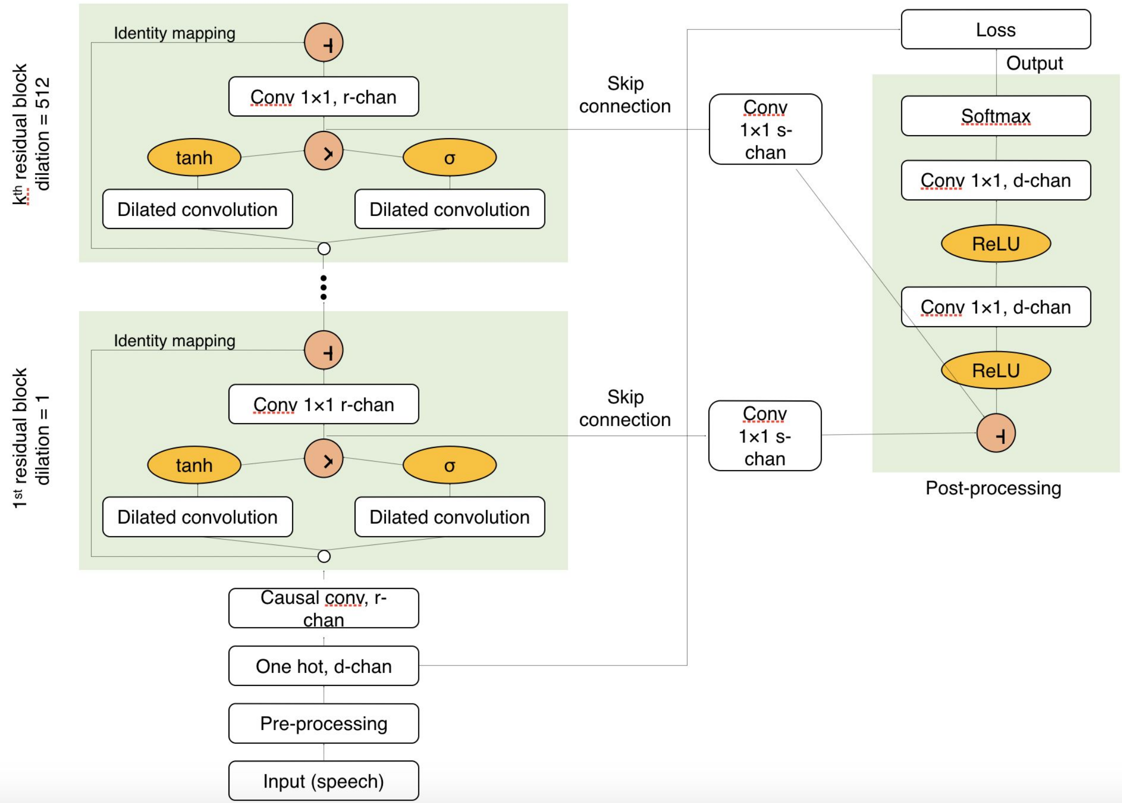
Wavenet – Gated activation units

- Input passes through dilated convolution
- Output split into two paths
- One path goes through tanh activation [-1 to 1] (Controls feature values)
- Other path goes through sigmoid activation [0 to 1](act like a gate)
- Results are multiplied element-wise
- Can effectively "turn off" irrelevant temporal information

$$z = \tanh(W_{f,k} * x) \odot \sigma(W_{g,k} * x)$$



WaveNet: Overall architecture



Data Flow Process

Input

- Raw audio waveform quantized to 256 possible values
- One-hot encoded input
- Conditional inputs (optional):
    - Speaker embeddings
    - Linguistic features

Hidden Layers

- Each residual block processes at current dilation level
- Information flows through:
    - Main path (residual connections)
    - Skip connections to final layers

Output Layer

- Combines skip connections
- ReLU activation
- 1x1 convolution
- Softmax to predict next sample

WaveNet: Training Process

Data Preparation

- Audio sampled at 16kHz
- Quantized to 8-bit mu-law encoding
- Segmented into training examples

Loss Function

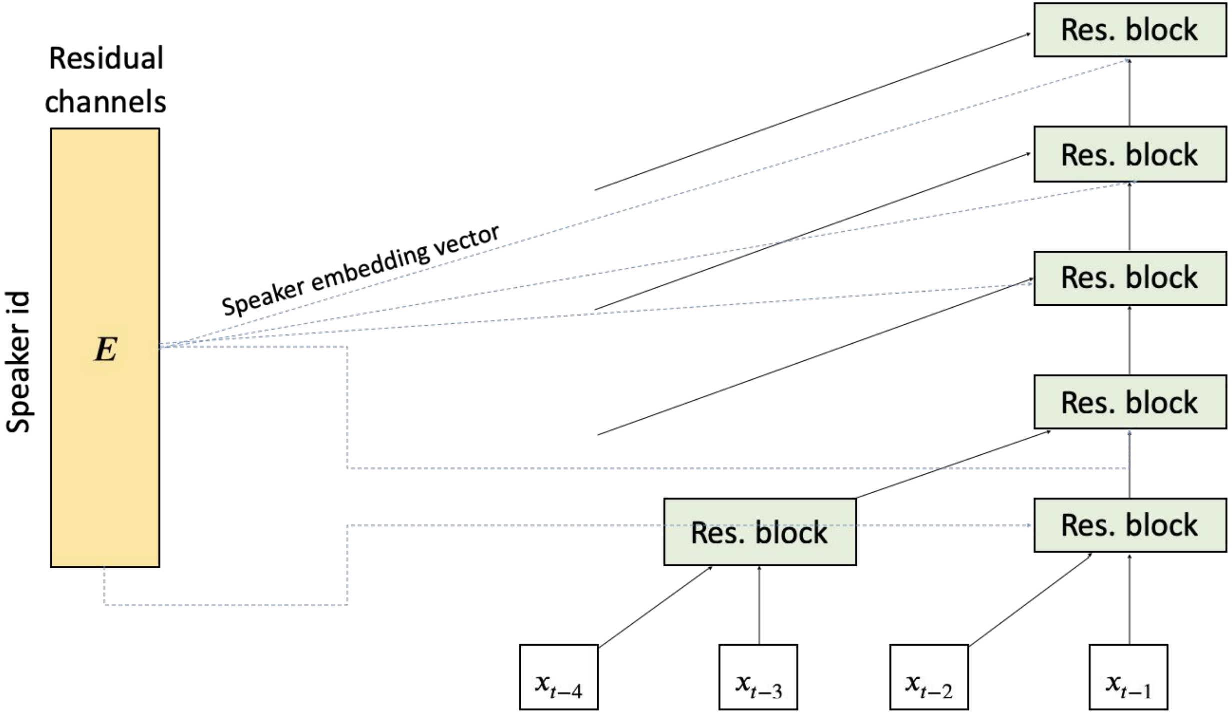
- Cross-entropy between predicted and actual next sample
- Optimized using Adam optimizer

Generation

- Autoregressive generation one sample at a time
- Each output feeds back as input
- Parallel generation using fast wavenet technique

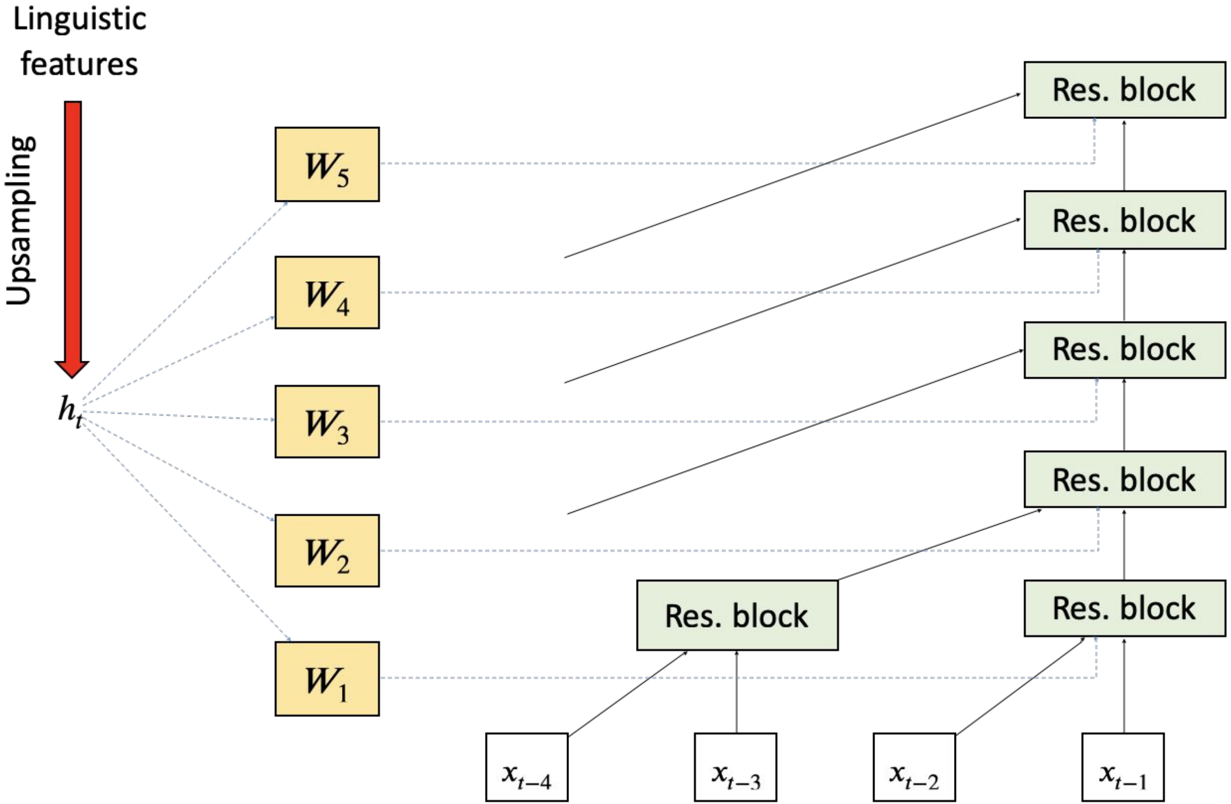
WaveNet architecture – Global conditioning

$$p\left(x_t \mid x_{t-R}, \ldots, x_{t-1}, g\right)$$



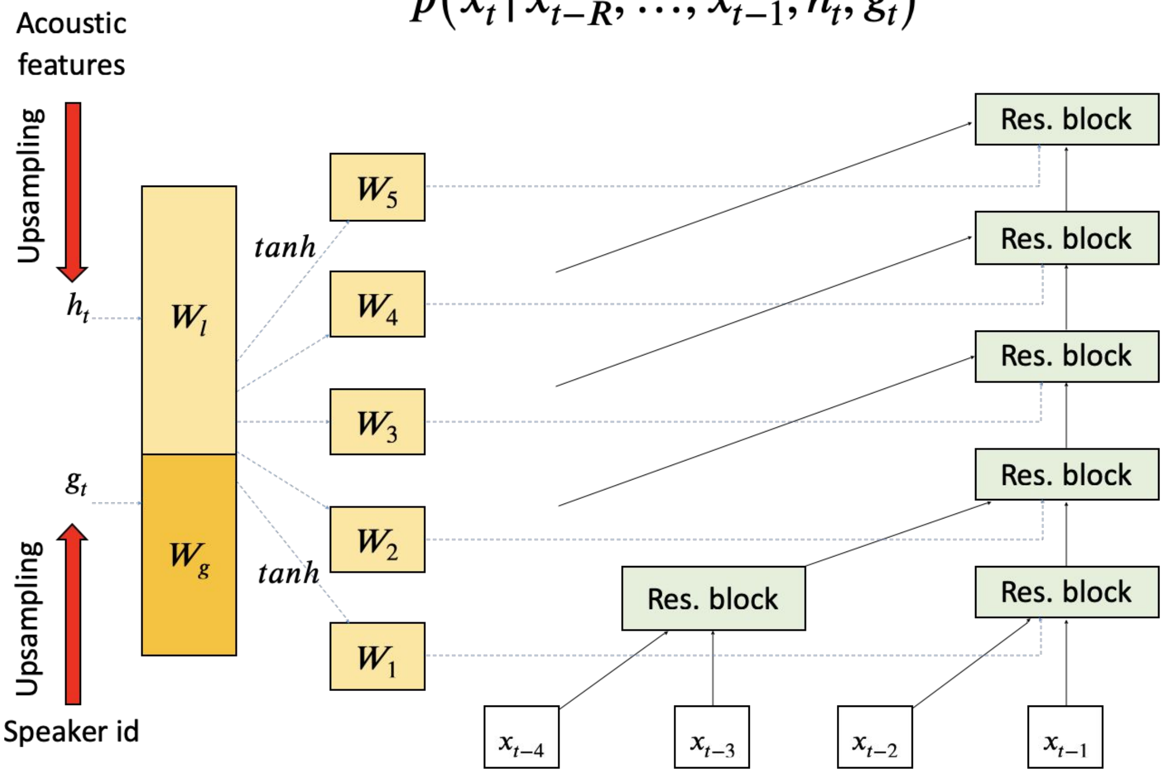
WaveNet architecture – Local conditioning

$$p\left(x_t \mid x_{t-R}, \ldots, x_{t-1}, h_t\right)$$

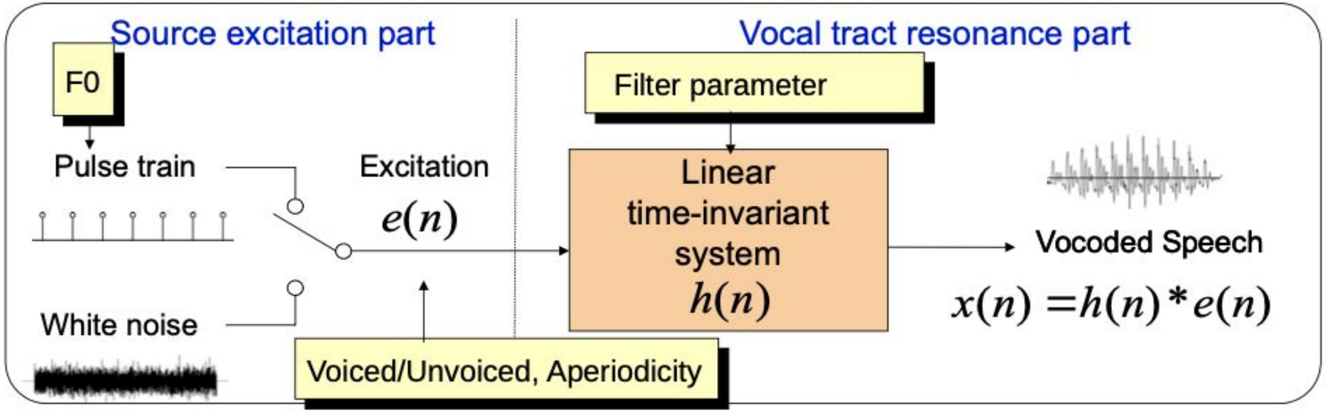


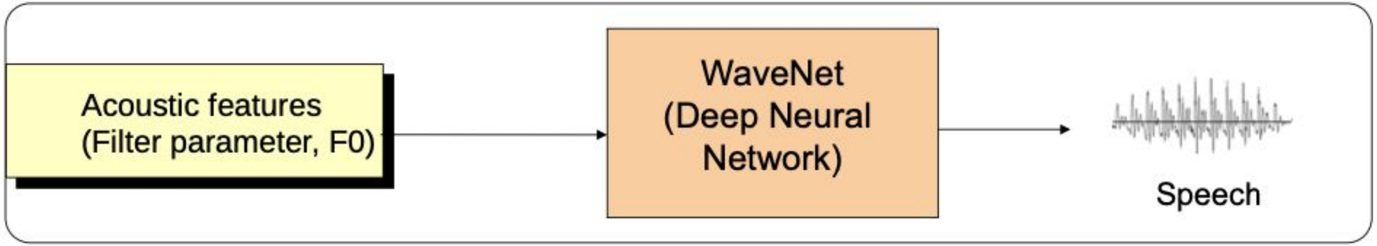
WaveNet architecture – Local and global conditioning

$$p\left(x_t \mid x_{t-R}, \ldots, x_{t-1}, h_t, g_t\right)$$



WaveNet Vocoder:





Adiga, Nagaraj, Vassilis Tsiaras, and Yannis Stylianou. "On the use of WaveNet as a statistical vocoder." *Proc. ICASSP*. IEEE, 2018.

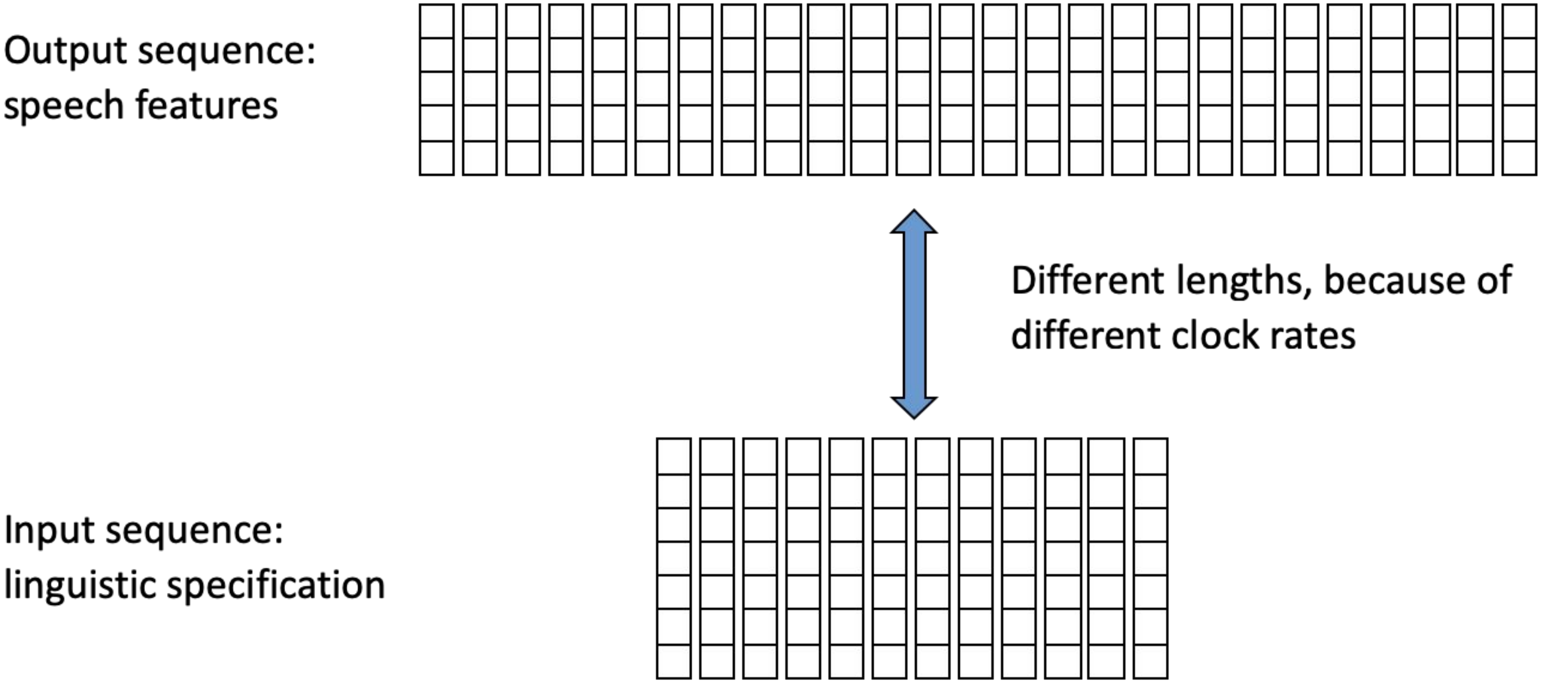
Tacotron 2: Text to acoustic features

- Input
  - Phoneme sequence
    - G2P model used to map text to phonemes
- Output
  - Mel-spectrogram
    - 80 bands
    - 50ms window, 12.5ms hop, Hanning window @ 16 kHz sampling rate
- Training:
  - Used Encoder-decoder architecture

Shen, Jonathan, et al. "Natural tts synthesis by conditioning wavenet on mel spectrogram predictions." IEEE, Proc. I*CASSP,* 2018.

Sequence-to-sequence regression

The core problem can be described as sequence-to-sequence regression.



Tacotron 2: Encoder-Decoder architecture

Encoder:

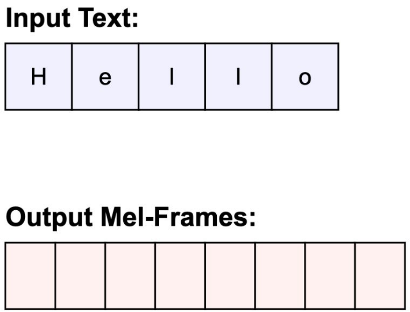
- Character embeddings (512 dimensions)
- 3 Convolutional layers (5x1 kernel, 512 channels): Feature extraction from text
- Batch normalization and ReLU
- Bi-directional LSTM (256 units per direction): Captures long-term dependencies in both directions and helps understand full sentence context

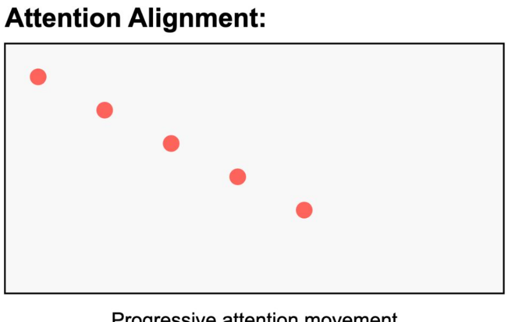
Decoder:

- Location-sensitive attention
- 2 LSTM layers (1024 units): Generates mel-spectrogram frames autoregressively
- PreNet: 2 dense layers with dropout, act like information bottleneck
- PostNet: Refines the generated mel-spectrogram predictions
- Stop token prediction

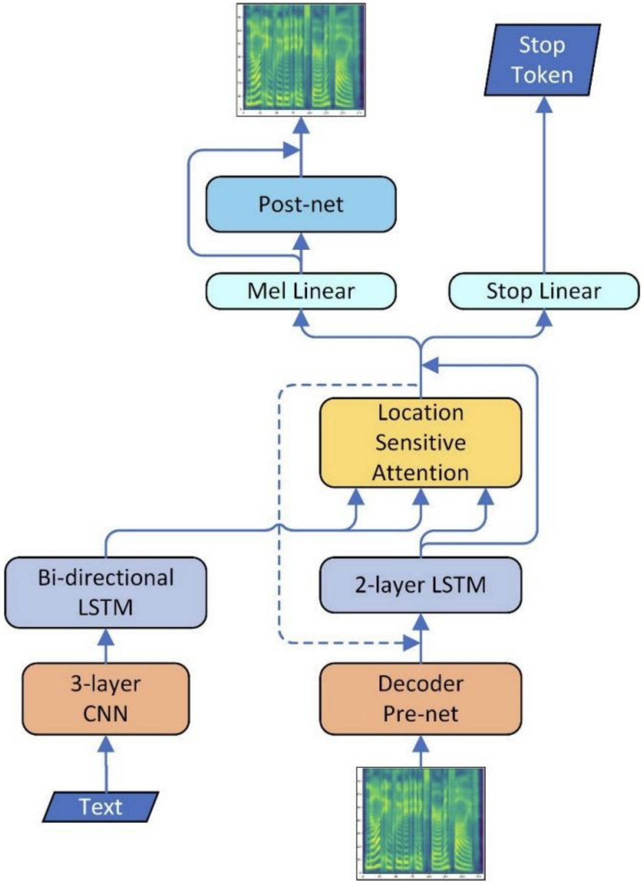
Why attention is needed?

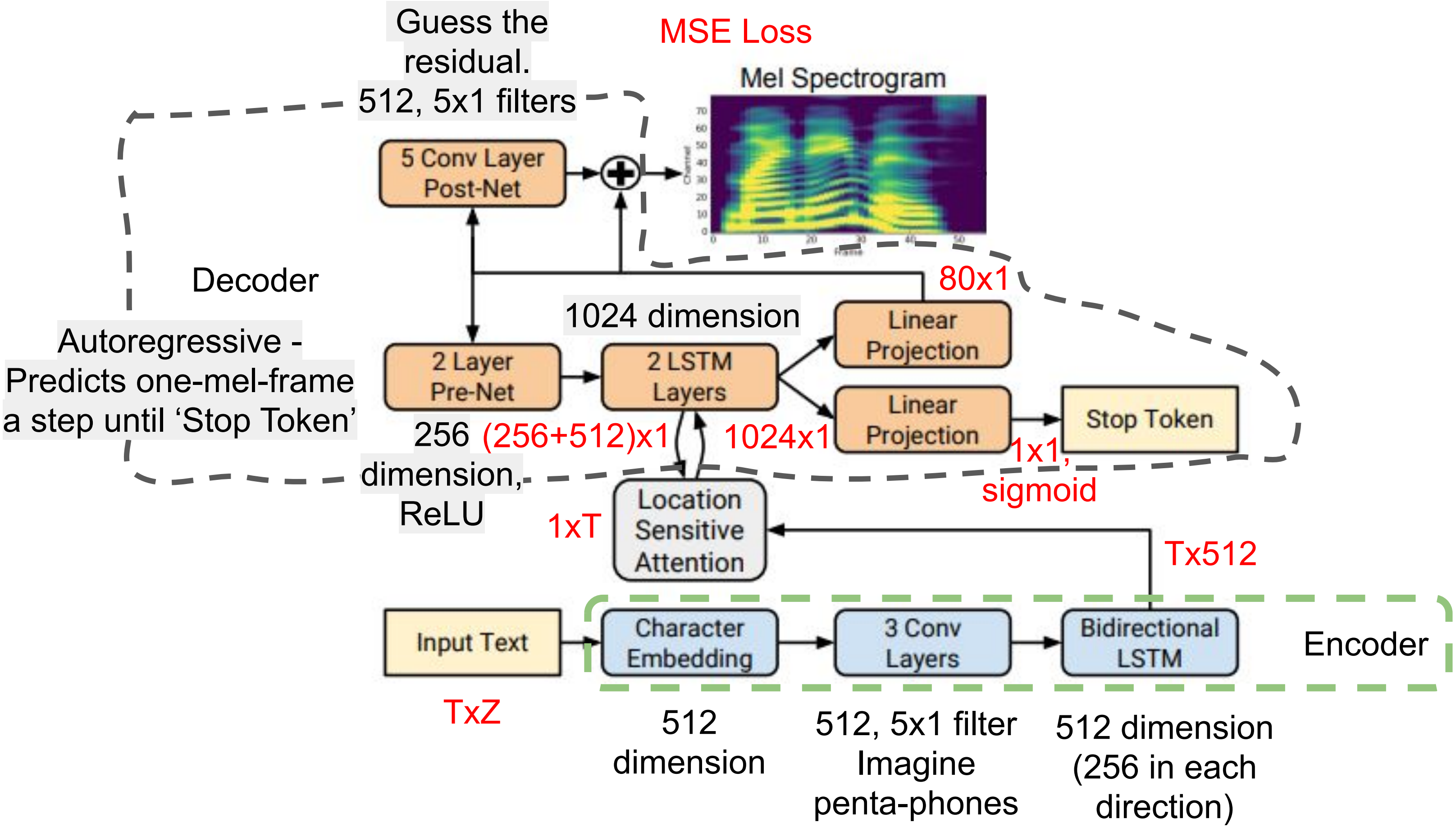
- Text to speech is not a one-to-one mapping
- Single character can produce multiple audio frames and duration varies based on context
- Query: decoder output, keys: Encoder input, Values: Encoder input
- Compute attention scores: energy = torch.bmm(query, keys.transpose(1, 2))
- Normalize attention weights: attention_weights = torch.softmax(energy, dim=-1)
- Compute context vector: context = torch.bmm(attention_weights, values)
- For one decoder frame: attention_row = [0.7, 0.2, 0.05, 0.03, 0.02] Focuses mainly on first encoder state





Tacotron 2: Block diagram





Guess the residual.
512, 5x1 filters
MSE Loss
Mel Spectrogram
5 Conv Layer Post-Net
Decoder
80x1
1024 dimension
Linear Projection
Autoregressive - Predicts one-mel-frame a step until 'Stop Token'
2 Layer Pre-Net
2 LSTM Layers
Linear Projection
Stop Token
256 dimension, ReLU
(256+512)x1
1024x1
1x1, sigmoid
1xT
Location Sensitive Attention
Tx512
Input Text
Character Embedding
3 Conv Layers
Bidirectional LSTM
Encoder
TxZ
512 dimension
512, 5x1 filter
Imagine penta-phones
512 dimension
(256 in each direction)

Tacotron 2: Training

- Dataset
  - Proprietary US English dataset
  - 24.6 hours of speech
- Training:
  - Teacher forcing during training
  - Multiple loss components: [loss = mel_loss + stop_token_loss + attention_loss]
  - Gradual reduction of teacher forcing ratio

Tacotron 2 - Audio samples from "Natural TTS Synthesis by Conditioning WaveNet on Mel Spectrogram Predictions"

	System	MOS
1025-dimensions	Tacotron 2 (Linear + G-L)	3.944 ± 0.091
	Tacotron 2 (Linear + WaveNet)	4.510 ± 0.054
80-dimensional	Tacotron 2 (Mel + WaveNet)	$\mathbf{4.526 \pm 0.066}$

System	MOS
Parametric	3.492 ± 0.096
Tacotron (Griffin-Lim)	4.001 ± 0.087
Concatenative	4.166 ± 0.091
WaveNet (Linguistic)	4.341 ± 0.051
Ground truth	4.582 ± 0.053
Tacotron 2 (this paper)	$\mathbf{4.526 \pm 0.066}$

References

- Klatt; "The Klatt-Talk text-to-speech system," in Proc. ICASSP, '82.
- Hunt, A. and Black, A. (1996). Unit selection in a concatenative speech synthesis system using a large speech database, in Proc. ICASSP 96, pp 373-376, Atlanta, Georgia.
- K. Tokuda, Y. Nankaku, T. Toda, H. Zen, J. Yamagishi, and K. Oura, “Speech synthesis based on hidden markov models", Proceedings of the IEEE, vol. 101-5, pp. 12341252, 2013.
- H. Zen, K. Tokuda, and A. W. Black, Statistical parametric speech synthesis, Speech Communication, vol.51(11), pp. 10391064, 2009
- Simon King, An introduction to statistical parametric speech synthesis, Sadhana Vol. 36, Part 5, October 2011, pp. 837{852
- Tokuda, K., Yoshimura, T., Masuko, T., Kobayashi, T., Kitamura, T. Speech parameter generation algorithms for HMM-based speech synthesis" in proceedings of ICASSP 2000.

References

- Xie, Tianxin, et al. "Towards Controllable Speech Synthesis in the Era of Large Language Models: A Survey." *arXiv preprint arXiv:2412.06602* (2024).
- X. Tan, T. Qin, F. Soong, and T.-Y. Liu, “A survey on neural speech synthesis,” arXiv preprint arXiv:2106.15561, 2021
- Triantafyllopoulos, Andreas, et al. "An overview of affective speech synthesis and conversion in the deep learning era." *Proceedings of the IEEE* 111.10 (2023): 1355-1381.
- Zhang, Chenshuang, et al. "A survey on audio diffusion models: Text to speech synthesis and enhancement in generative ai." *arXiv preprint arXiv:2303.13336* (2023).
- van den Oord, Aaron; Dieleman, Sander; Zen, Heiga; Simonyan, Karen; Vinyals, Oriol; Graves, Alex; Kalchbrenner, Nal; Senior, Andrew; Kavukcuoglu, Koray, WaveNet: A Generative Model for Raw Audio, arXiv:1609.03499, 2016
- Adiga, Nagaraj, Vassilis Tsiaras, and Yannis Stylianou. "On the use of WaveNet as a statistical vocoder." *Proc. ICASSP*. IEEE, 2018.
- Shen, Jonathan, et al. "Natural tts synthesis by conditioning wavenet on mel spectrogram predictions." IEEE, Proc. I*CASSP,* 2018.

References

- ISCA Synthesis Interest Group: https://synsig.org/index.php/Main_Page (look there for the SPCC videos)
- CSTR, Simon Kings' page on speech synthesis: http://www.speech.zone/courses/speech-synthesis/
- CMU Festvox: http://festvox.org/
- HTS web page http://hts.sp.nitech.ac.jp/ for more information on HMM based speech synthesis system (HTS)
- Google: https://google.github.io/tacotron/
- Espnet toolkit link: https://github.com/espnet/espnet

Thank You